

Jellyfin - Server



Introduction

This project demonstrates the deployment of a Jellyfin Media Server on an Ubuntu virtual machine hosted in Proxmox VE. Secure remote connectivity is provided using ZeroTier, while Containerlab is used to create and test a containerized Linux environment.

Key Features:

- Virtualized server environment.
- Secure remote access.
- Container-based networking.
- Practical networking tests.

Project Objectives

- Deploy Jellyfin on Ubuntu.
- Host the server using Proxmox VE.
- Configure secure remote access with ZeroTier.
- Build a simple Containerlab topology.
- Verify network connectivity.

Technologies Used

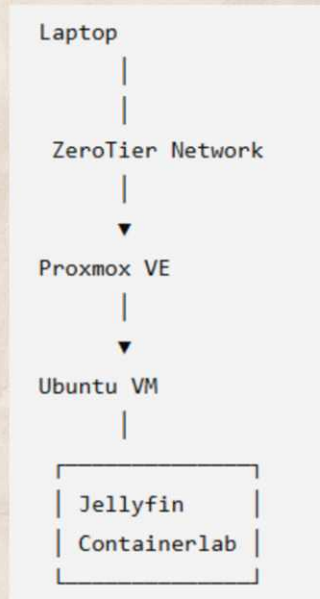
Technology :

- Proxmox VE
- Ubuntu Server
- Jellyfin
- ZeroTier
- Docker
- Containerlab

Purpose :

- Virtualization Platform
- Operating System
- Media Server
- Secure Virtual Network
- Container Engine
- Network Laboratory

System Architecture



Implementation Steps

1. Installed Proxmox VE.
2. Created Ubuntu Virtual Machine.
3. Installed Jellyfin.
4. Installed ZeroTier.
5. Authorized the device.
6. Installed Docker.
7. Installed Containerlab.
8. Created the lab topology.
9. Tested connectivity.


```
mediaadmin@jellyfin-server:~$ containerlab version

  ++gg6++qg++ ++++ ^g*^g  ++++ ++ ++++ ++ggb*^gg++ ++ ++
  ++ ++ ++++ ++++ ++++ ++ ++++ ++++ ++++ ++++ ++++ 1 ++ "qg++*^g++
  ^++ + ++++ ++b++ ^++ ++ ++(7++ ++ ++ ++ ++c++ ++ ++
  @dggc cggfa ^g @g ^b jb ^g gb gb db ggggb^g dh gb@ggdb gggga

version: 0.77.0
commit: 05c2006e3
date: 2026-06-28T08:07:27Z
source: https://github.com/srl-labs/containerlab
rel. notes: https://containerlab.dev/rn/0.77/
mediaadmin@jellyfin-server:~$
```

```
mediaadmin@jellyfin-server:~/clab-demo$ sudo docker exec -it clab-demo-linux1 sh
/ # ping -c 3 192.168.68.104
PING 192.168.68.104 (192.168.68.104): 56 data bytes
64 bytes from 192.168.68.104: seq=0 ttl=63 time=154.524 ms
64 bytes from 192.168.68.104: seq=1 ttl=63 time=60.453 ms
64 bytes from 192.168.68.104: seq=2 ttl=63 time=81.628 ms

--- 192.168.68.104 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 60.453/98.868/154.524 ms
```


Challenges

- ZeroTier authorization.
- Docker permission issue.
- Container deployment.
- Remote access configuration.

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Solutions

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- Used sudo with Docker.
- Redeployed the Containerlab topology.
- Verified connectivity using Ping.

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Conclusion

The project successfully demonstrated the deployment of a complete virtual media server environment. Jellyfin was hosted on an Ubuntu virtual machine using Proxmox VE, securely connected through ZeroTier, and integrated with Containerlab for network testing.



References

- <https://jellyfin.org>
- <https://www.proxmox.com>
- <https://www.zerotier.com>
- <https://containerlab.dev>
- <https://docs.docker.com>





Our Team

Maya

Huda



ce



Thank
You

